Introduction to Logic: Lecture 2

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October 27 2026

What this course is about (**reminder**)...

Logic: the study of reasoning, inference, and how one constructs correct arguments.

An argument in English

"You can access the Internet from campus only if **you are a linguistics major** or you are **not a freshman**. Mary **can access the Internet**. So, Mary is **a linguistics major** or **not a freshman**."

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We will model arguments using symbolic logic

$$P_a \rightarrow (P_l \vee \neg P_f), \quad P_a \quad \therefore \quad P_l \vee \neg P_f$$

Tools we will learn along the way and **where we are**

Understanding and building logics requires a few mathematical tools:

$\{a, b, c\}$

Sets

linguistics majors

$R(a, b)$

Relations

loves, older than

$f : A \to B$

Functions

capital of

$\vdash$

Proofs

informal & formal

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last lecture

Last lecture: set theory

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Formal analysis requires talking about *groups of things*:

L	F	A
linguistics majors	freshmen	Internet users

Set theory: the mathematical study of **sets**, i.e., collections of distinct objects; how to compare sets, build new ones.

Tools we will learn along the way and **where we are**

Understanding and building logics requires a few mathematical tools:

$\{a, b, c\}$	$R(a, b)$	$f : A \to B$	$\vdash$
Sets	**Relations**	**Functions**	**Proofs**
linguistics majors	*loves, older than*	*capital of*	informal & formal
✓ last lecture	today	today	later

From sets to relations

An argument

"You can access the Internet from campus only if you are a linguistics major or you are not a freshman. **Mary can access the Internet**. So, Mary is a linguistics major or not a freshman."

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A set: a property of *one* thing

Mary is a linguistics major

$\text{Mary} \in L$

A relation: connects *two* things

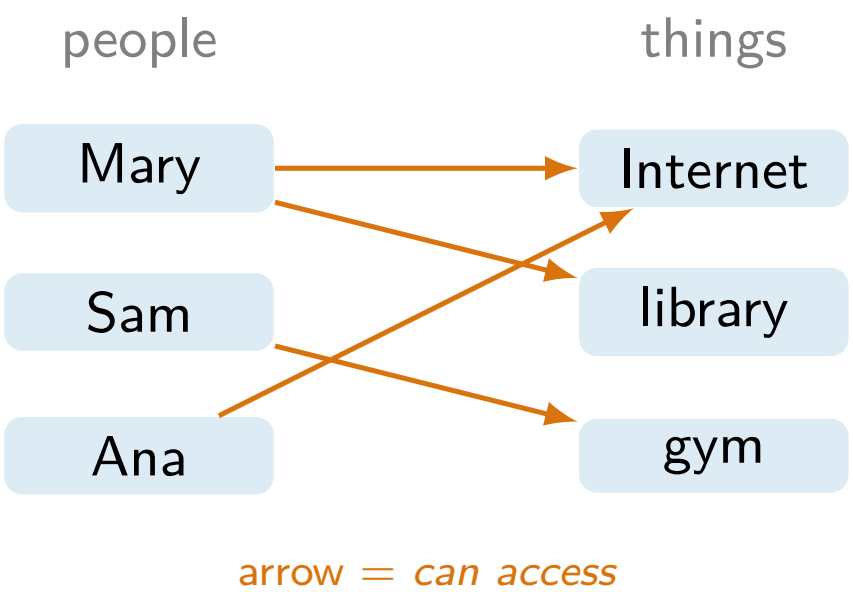
Mary can access the Internet

$\textsc{Access}(\text{Mary}, \text{Internet})$

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Who can access what?

ACCESS(Mary, Internet)

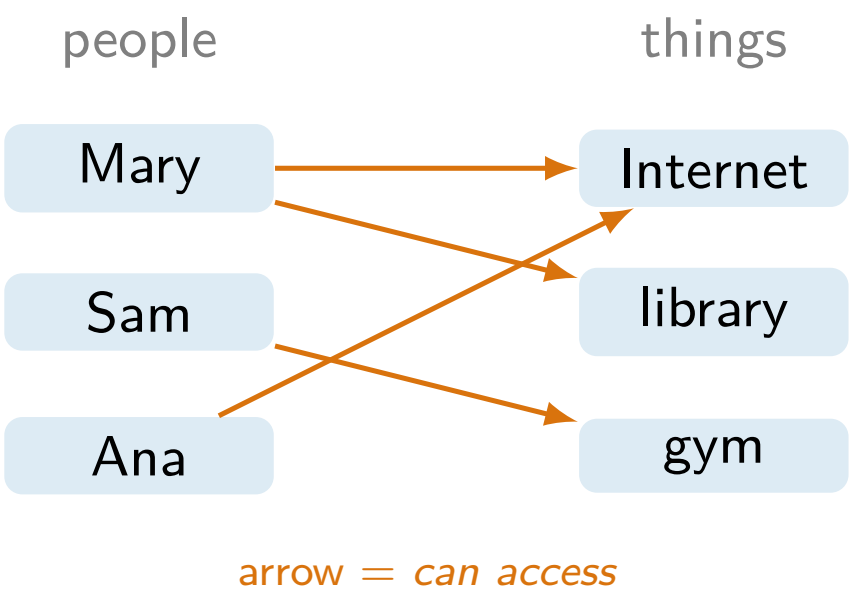
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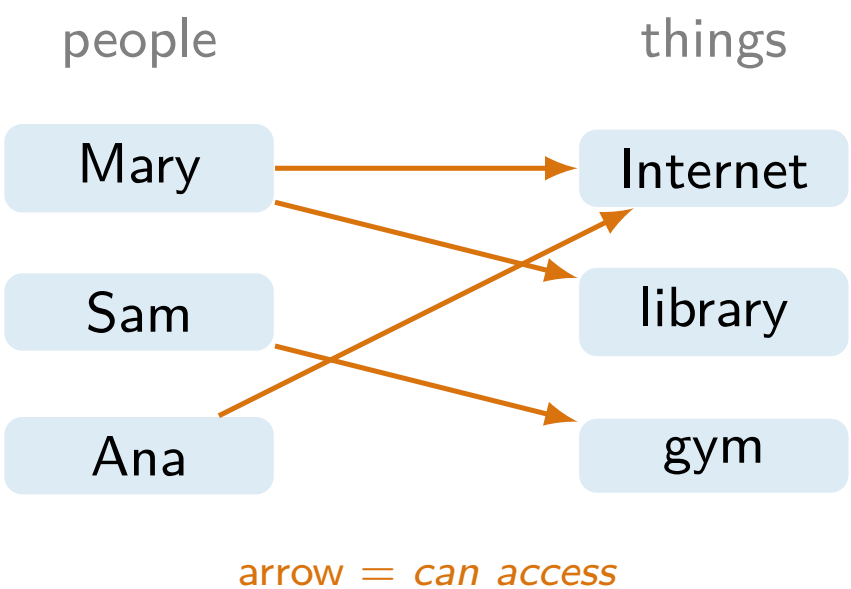
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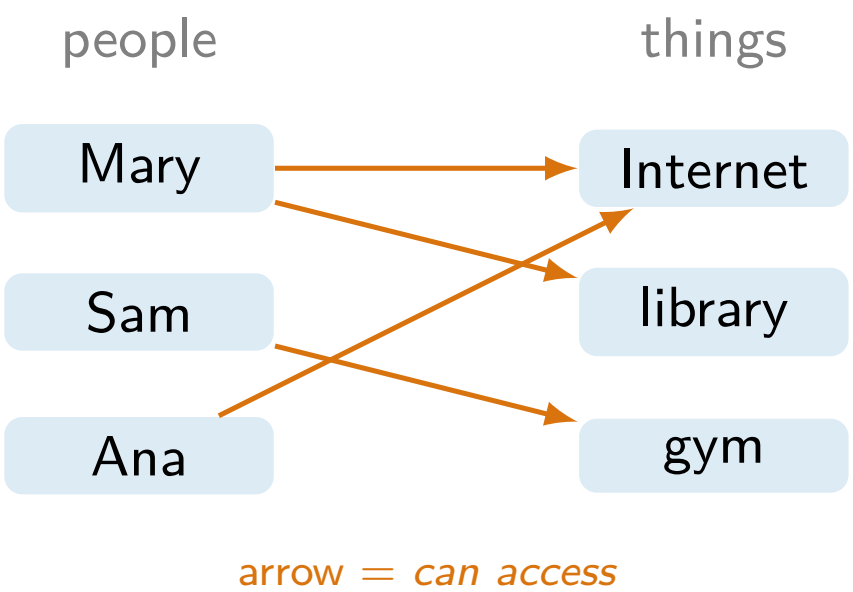
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Relations describe how things are *connected*, we will show how they can be formalized as types of sets.

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Functions: a special kind of relation where each input has *exactly one* output. We will cover these too.

Set theory reminders

What we covered, what you should know

Defining sets

- ✓ notation: $\{\ldots\}$, $\in$, $\notin$, $\emptyset$, U
- ✓ $\{a, b\} = \{b, a\} = \{a, a, b\}$
- ✓ listing, set-builder, recursion

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Comparing sets

- ✓ check $X \subseteq Y$ and $X \subset Y$
- ✓ find a counterexample when not
- ✓ $\emptyset \subseteq X$ for every X
- ✓ read and draw Venn diagrams

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Operations

- ✓ $X \cap Y$, $X \cup Y$, $X \setminus Y$, $\overline{X}$
- ✓ $\mathcal{P}(X)$ and $|\mathcal{P}(X)| = 2^{|X|}$

Comprehension check: the empty set

True or false?

1. $\emptyset \in \emptyset$
2. $\emptyset \subseteq \emptyset$
3. $|\{\emptyset\}| = 0$
4. $\emptyset \in \{\emptyset\}$
5. $\mathcal{P}(\emptyset) = \emptyset$
6. $|\mathcal{P}(\mathcal{P}(\emptyset))| = 2$

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6.	$\lvert\mathcal{P}(\mathcal{P}(\emptyset))\rvert = 2$	**true**	$\mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$

Some of these are tricky!

Comprehension check: $\in$, $\subseteq$, and operations

True or false?

1. $\{a\} \in \{a, b\}$
2. $\{a\} \subseteq \{\{a\}, b\}$
3. $\{a\} \in \mathcal{P}(\{a, b\})$
4. $|\mathcal{P}(\{a, b, c\})| = 8$
5. $|\{a, \{b\}\} \cup \{b\}| = 2$
6. $X \setminus Y = Y \setminus X$, always

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5. $|\{a, \{b\}\} \cup \{b\}| = 2$ — **false** — $\{a, \{b\}, b\}$: $b \neq \{b\}$
6. $X \setminus Y = Y \setminus X$, always — **false** — counterexample: $\{a\} \setminus \emptyset \neq \emptyset \setminus \{a\}$

Questions?

The **naïve** part in naïve set theory

Can a set contain itself?

Sets can contain sets. So can a set be *one of its own elements*?

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Naïve set theory allows *any* property to define a set. So consider the property $x \notin x$: "x does not contain itself".

An amusing paradox: **Russell's paradox**

The set of all sets that do not contain themselves

$$R = \{x \mid x \notin x\}$$

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Lesson: our *naïve* idea of a set (any property defines a set) leads to fundamental problems. Modern set theories add careful rules to avoid this. **This result had a big impact on the foundations of mathematics!**

Relations

Fundamentals: tuples and ordered pairs

Sets: no order

$$\{x, y\} = \{y, x\}$$

cities Hans visited last summer:

$$\{\text{London}, \text{Paris}, \text{Berlin}\}$$

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Tuples: order matters

$$\langle x, y\rangle \neq \langle y, x\rangle \quad (\text{if } x \neq y)$$

. . . in the order he visited them:

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Writing tuples

$$\langle \underbrace{\text{Berlin}}_{\text{1st}}, \underbrace{\text{London}}_{\text{2nd}}, \underbrace{\text{Berlin}}_{\text{3rd}}, \underbrace{\text{Paris}}_{\text{4th}} \rangle$$

angle brackets $\langle \ldots \rangle$ instead of curly braces; every element has a *position*

Pairs, triples, . . . , n-tuples

Tuples are named by their number of positions

$\langle x, y \rangle$	an **ordered pair** (2 positions)
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Two tuples are **equal** when they agree *position by position*:

$$\langle a, b \rangle = \langle c, d \rangle \quad \text{exactly when} \quad a = c \text{ and } b = d$$

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Compare with sets

$\{1, 2\} = \{2, 1\}$	$\langle 1, 2 \rangle \neq \langle 2, 1 \rangle$	order matters
$\{a, a, b\} = \{a, b\}$	$\langle a, a, b \rangle \neq \langle a, b \rangle$	repetition matters

Tuples can be built out of sets

In mathematics, virtually everything is built from sets, including tuples. A common definition:

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Try it: $\langle a, a \rangle = ?$

$\{\{a\}, \{a, a\}\}$

$= \{\{a\}, \{a\}\} = \{\{a\}\}$

still works: only a is there, in first position.

Fundamentals: the Cartesian product

The **Cartesian product** of X and Y is the set of *all* pairs with a first element from X and a second from Y:

$$X \times Y = \left\{ \langle x, y \rangle \mid x \in X,\ y \in Y \right\}$$

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How to build $X \times Y$

1. take each $x \in X$ in turn
2. pair it with *every* $y \in Y$
3. collect pairs, make a set.

$$X = \{a, b\}, \qquad Y = \{c, d\}$$

c d

a

b

Fundamentals: the Cartesian product

The **Cartesian product** of X and Y is the set of *all* pairs with a first element from X and a second from Y:

$$X \times Y = \left\{ \langle x, y \rangle \mid x \in X,\ y \in Y \right\}$$

How to build $X \times Y$

1. take each $x \in X$ in turn
2. pair it with *every* $y \in Y$
3. collect pairs, make a set.

$$X = \{a, b\}, \qquad Y = \{c, d\}$$

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$$X \times Y = \{\langle a, c \rangle, \langle a, d \rangle, \langle b, c \rangle, \langle b, d \rangle\} \quad 2 \times 2 = 4 \text{ pairs}$$

More Cartesian products

Still with $X = \{a, b\}$ and $Y = \{c, d\}$:

A set with itself

$X \times X = \{\langle a, a\rangle, \langle a, b\rangle, \langle b, a\rangle, \langle b, b\rangle\}$

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More than two sets: triples

$X \times X \times Y = \{\langle a, a, c\rangle, \langle a, b, c\rangle, \langle b, a, c\rangle, \ldots, \langle b, b, d\rangle\}$ $2 \times 2 \times 2 = 8$

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Shorthand

$X^n = \underbrace{X \times X \times \cdots \times X}_{n \text{ times}}$ e.g., $X^2 = X \times X$

Cartesian products are everywhere

A chessboard

$\{a, \ldots, h\} \times \{1, \ldots, 8\}$

each square is a pair: $\langle e, 4 \rangle$ is "e4"; 64 squares

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Word combinations

$\{\mathit{the}, \mathit{a}\} \times \{\mathit{cat}, \mathit{dog}\}$

each determiner with each noun: $\langle \mathit{the}, \mathit{cat} \rangle, \langle \mathit{a}, \mathit{dog} \rangle, \ldots$

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Quick facts

$|X \times Y| = |X| \cdot |Y|$ rows $\times$ columns in the grid

$X \times \emptyset = \emptyset$ nothing to pair with

What is a relation?

A **relation** between X and Y is a *subset* of their Cartesian product:

$$R \subseteq X \times Y$$

If $\langle x, y \rangle \in R$, we say that x *stands in relation* R to y.

This is a *binary* relation: it relates *two* things.

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1. Start from all pairs

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b	$\langle b, c \rangle$	$\langle b, d \rangle$

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1. Pick some pairs

	c	d
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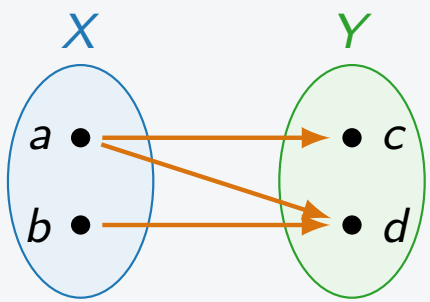
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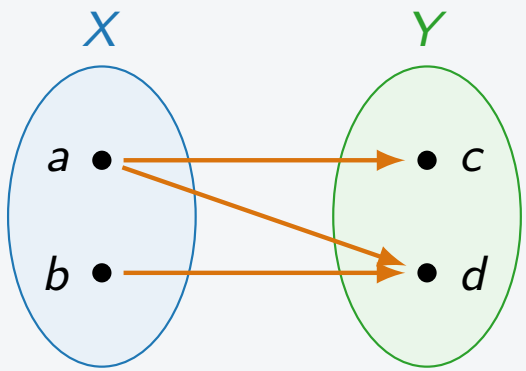
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2. Draw each pair as an arrow



three pairs, three arrows

Visualizing binary relations: arrow diagrams



oval	a set, e.g., X
dot	an element
arrow	a pair in R: $\langle a, c \rangle \in R$
no arrow	not in R: $\langle b, c \rangle \notin R$
all arrows	all of $X \times Y$

Like a Venn diagram, but with arrows *between* elements.

Relations on a single set

X and Y can be the *same* set: $R \subseteq X \times X$ is a **relation on** X.

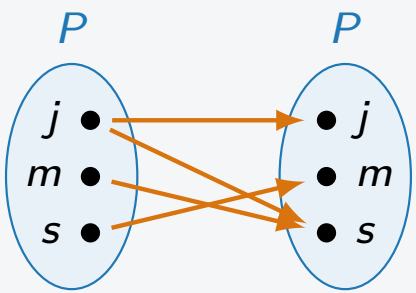
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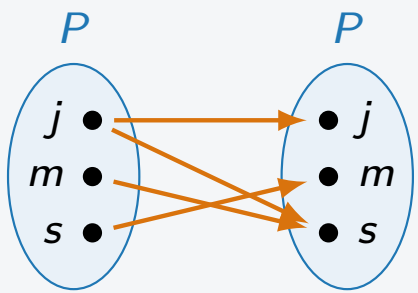


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works, but repetitive

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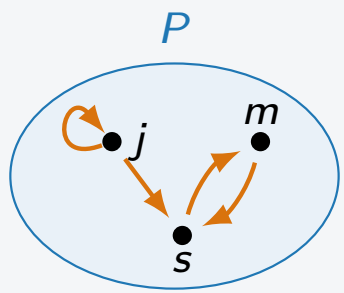
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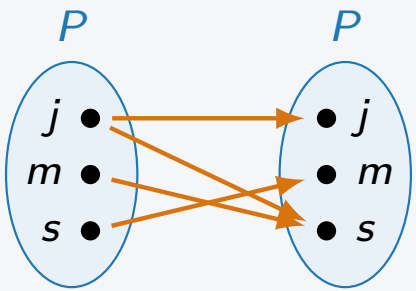
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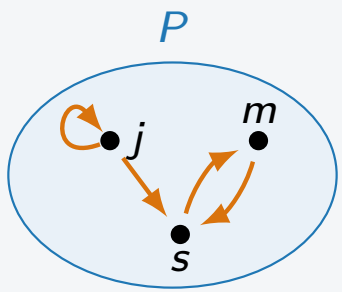
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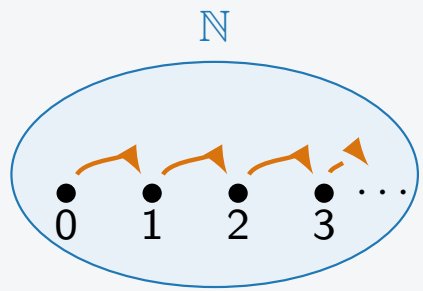
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Pred $\subseteq \mathbb{N} \times \mathbb{N}$

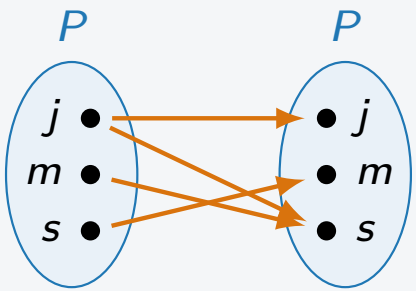


$\{\langle 0, 1 \rangle, \langle 1, 2 \rangle, \ldots\}$
is the predecessor of

Relations on a single set

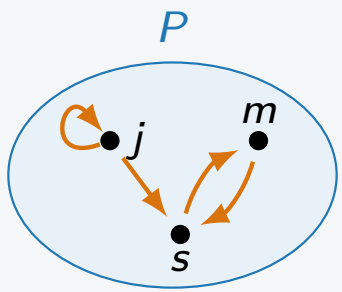
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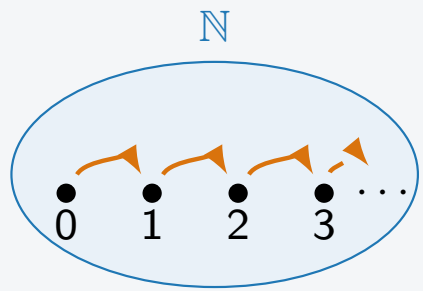
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Many familiar relations are relations on one set

$<$, $\leq$, $=$ on $\mathbb{N}$ *is older than*, *is a sibling of* on *People*

Relations: notation and special cases

Notation: shorter ways to write $\langle x, y \rangle \in R$

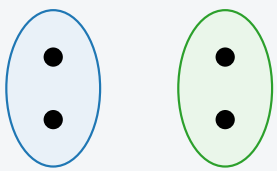
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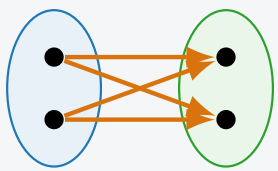
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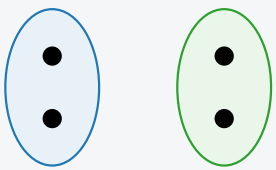
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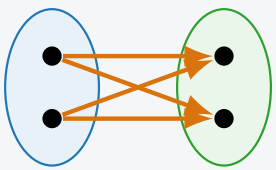
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How many relations between X and Y?

one for each subset of $X \times Y$:

$$|\mathcal{P}(X \times Y)| = 2^{|X| \cdot |Y|}$$

for $X = \{a, b\}$, $Y = \{c, d\}$: $2^4 = 16$ relations

Generalizing: n-ary relations

An n-**ary relation** is a subset of a Cartesian product of n sets:

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so far: the case $n = 2$, *binary* relations

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Arrow diagrams only work for binary relations

An arrow has exactly *two* ends, so it can only show a pair. For $n \geq 3$, we list the tuples instead, e.g., as a table.

giver	gift	recipient
Mary	book	Sam
Sam	flowers	Ana

Relations are the foundation of databases

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-- who gave Sam something?
SELECT giver
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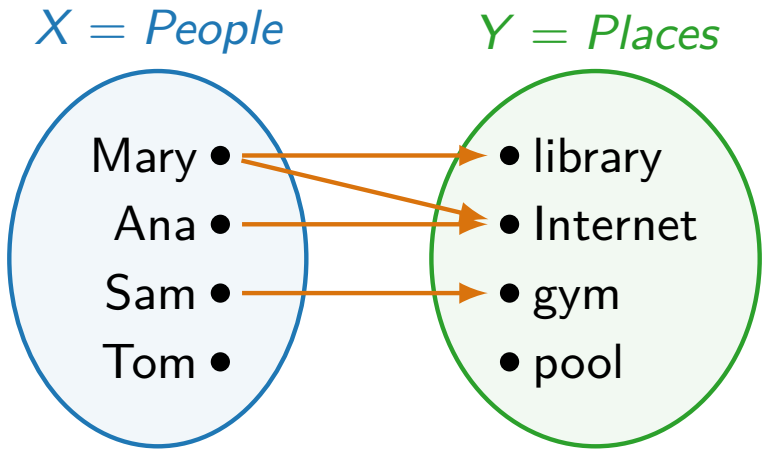
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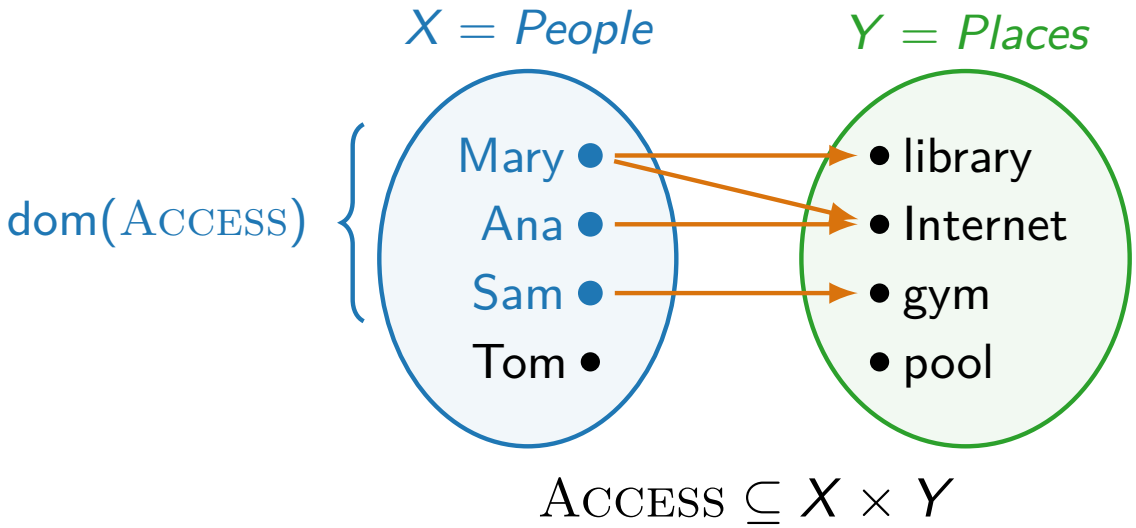
Booking a flight, checking your bank balance, scrolling a social media feed: behind each one, a program is querying relations, just like we did above.

Important: domain and range of a relation



$\textsc{Access} \subseteq X \times Y$

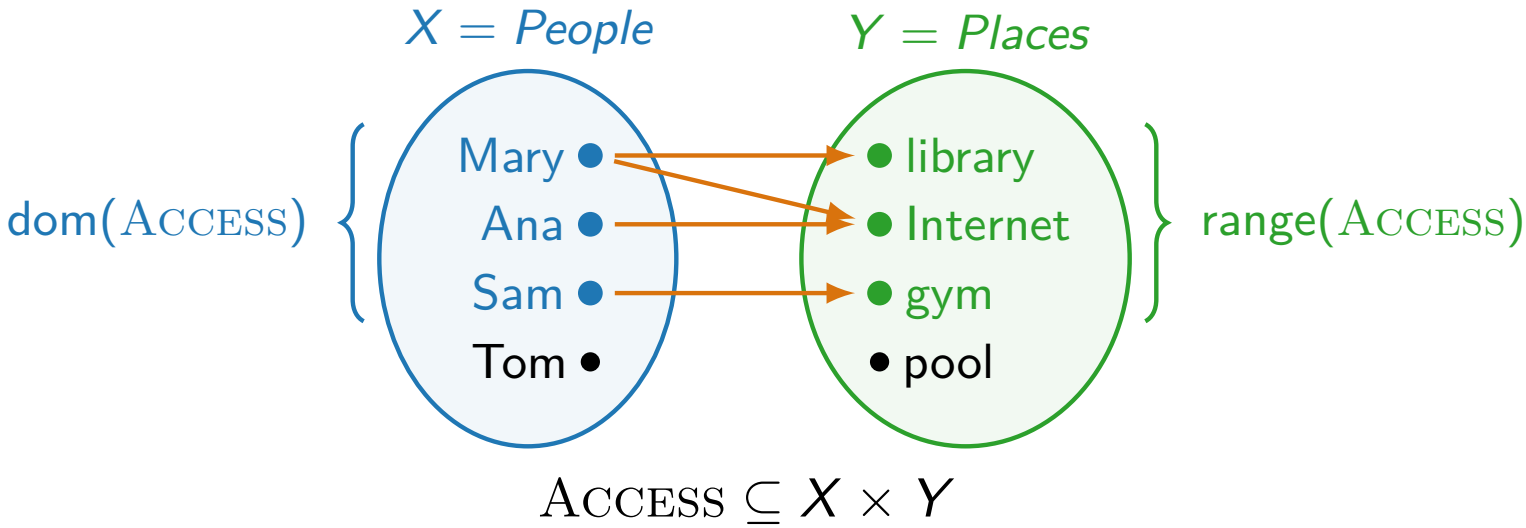
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Domain: the elements with an arrow going *out*

$\mathrm{dom}(R) = \{x \in X \mid Rxy \text{ for some } y\}$

Important: domain and range of a relation



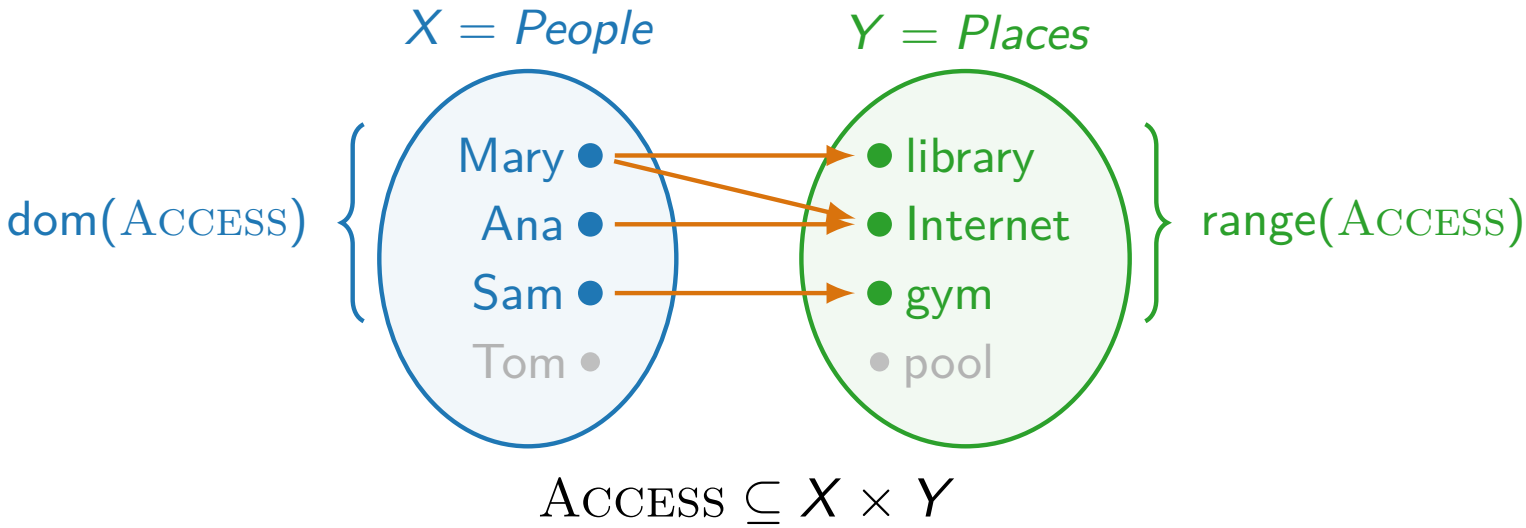
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Both can be smaller than X and Y, i.e., $\mathrm{dom}(R) \subseteq X$, $\mathrm{range}(R) \subseteq Y$.

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unlike sets, where order is ignored

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$$R \subseteq X \times Y, \quad \text{written } Rxy$$

loves, *has access to*, $<$

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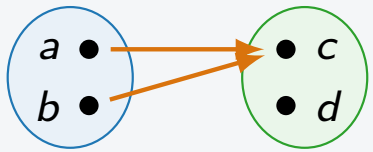
$$|X \times Y| = |X| \cdot |Y|$$

3. A relation: *some* of the pairs

$R \subseteq X \times Y$, written Rxy

loves, *has access to*, $<$

4. Arrow diagrams



one arrow per pair; *binary* only

Interim summary: tuples and relations

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$$\langle a, b\rangle \neq \langle b, a\rangle$$

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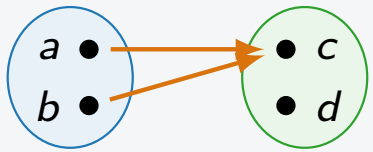
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$n = 1$: properties, $n = 3$: triples

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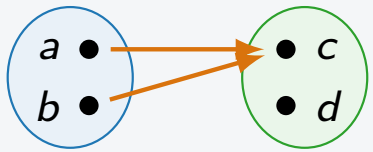
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6. Relations are everywhere

language, math, databases

a table is a relation, a row a tuple

Quick check: true or false?

1. $\langle 1, 2\rangle = \langle 2, 1\rangle$
2. $\{a, b\} \times \{c\}$ has two elements
3. $\langle c, a\rangle \in \{a, b\} \times \{c, d\}$
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5.	$\langle$Mary, book, Sam$\rangle$ is one arrow	**false**	arrows only show pairs

Questions?

Functions

From relations to functions: special kinds of relations

A **function** $f : X \to Y$ is a type of relation $f \subseteq X \times Y$ such that

1. every $x \in X$ has *at least one* arrow: $\quad \mathrm{dom}(f) = X$
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Together: every input is paired with **exactly one** output.

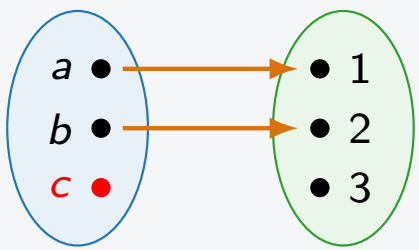
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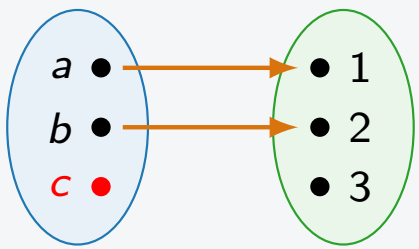
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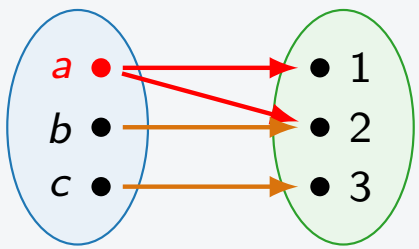
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a has *two* arrows:
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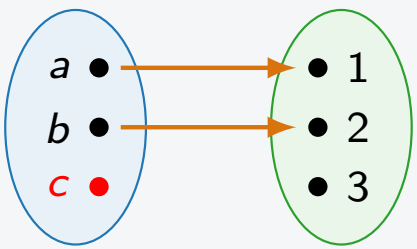
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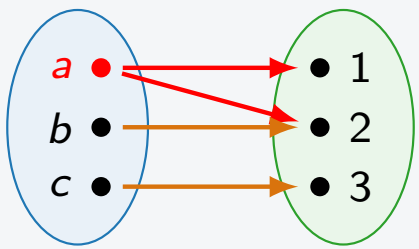
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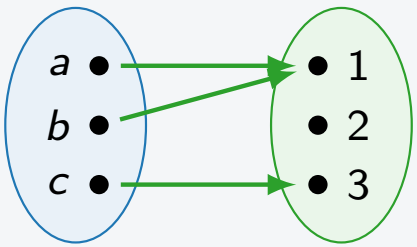
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✓ **a function**



exactly one arrow
out of each x

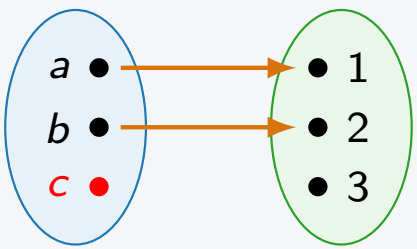
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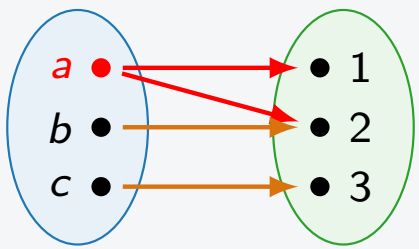
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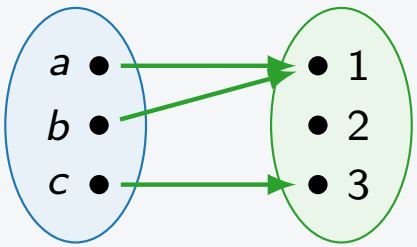
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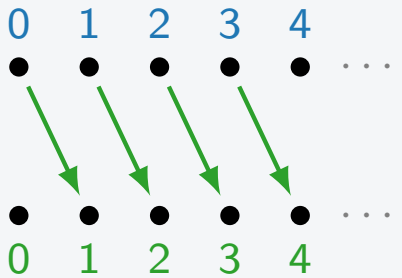
exactly one arrow
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Nothing is required of Y

a and b may share 1, and 2 may get no arrow: $\text{range}(f) \subseteq Y$. Y is called the **codomain**.

Functions on numbers: $f : \mathbb{N} \to \mathbb{N}$

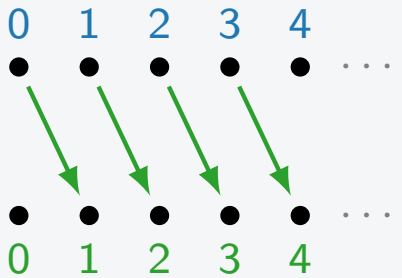
$f(x) = x + 1$



successor: this is PRED from before!

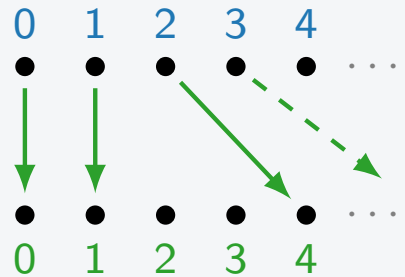
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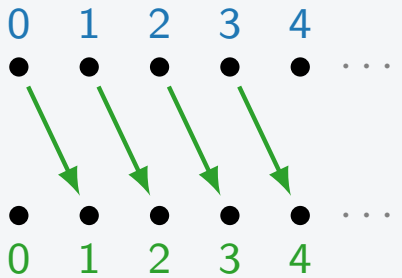


square: $f(3) = 9$

2, 3 never hit: fine

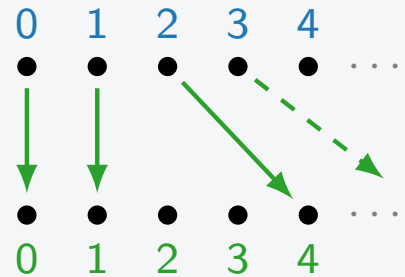
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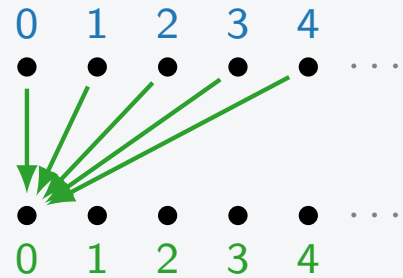
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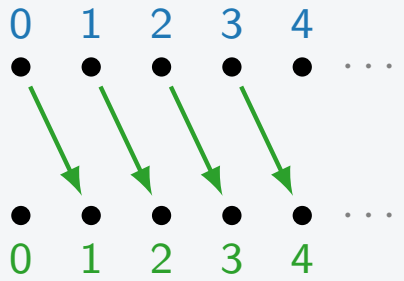
$f(x) = 0$



constant: all inputs share one output: fine

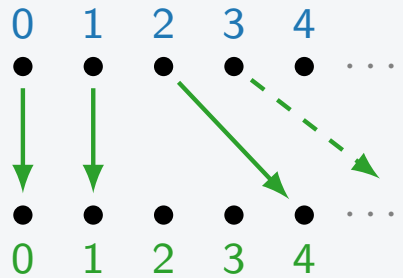
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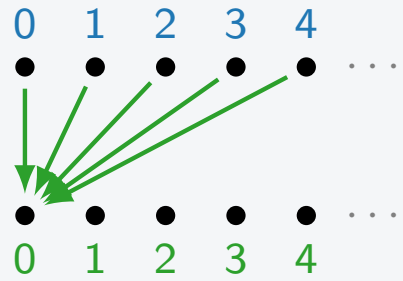
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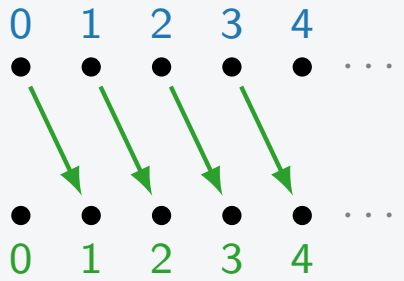
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$g(x) = x - 1$ $\quad$ $g(0) = -1$ is not in $\mathbb{N}$: 0 has no output (breaks 1.)

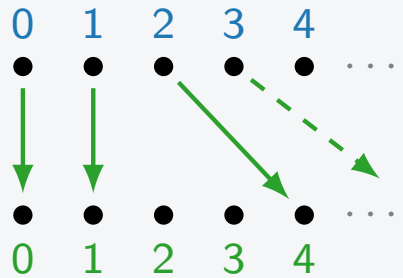
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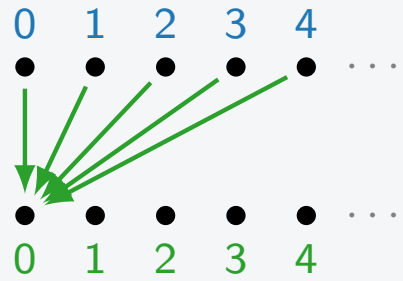
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the divisors of x	6 has four: $1, 2, 3, 6$ (breaks 2.)

Functions with more than one input

A function can take n inputs: $f : X_1 \times \cdots \times X_n \to Y$

Every n-tuple of inputs gets **exactly one** output, written $f(x_1, \ldots, x_n)$.

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Addition: $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$

$x \backslash y$	0	1	2
0	0	1	2
1	1	2	3
2	2	3	4

one value per cell of the grid;
$+(2, 1) = 3$ is the triple $\langle 2, 1, 3 \rangle$

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In language and data

$\textsc{Grade}(\text{Mary}, \text{Logic}) = \text{A}$

the price of x at store y

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Nothing new: the input set is just a Cartesian product

As a relation: $f \subseteq X_1 \times \cdots \times X_n \times Y$ an $(n{+}1)$-ary relation

Structural properties of relations and functions

Once we have a relation, we can ask what **shape** it has.



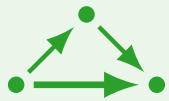
Reflexivity

where are the loops?



Symmetry

which arrows come back?



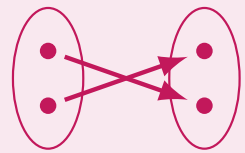
Transitivity

which paths have shortcuts?



Connected

are all pairs linked?



Functions

injective, surjective, bijective

relations on one set X: $R \subseteq X \times X$

$f : X \to Y$

Relational diagrams: relations on one set

For a relation on one set, $R \subseteq X \times X$, we drop the ovals: just dots and arrows.



dot

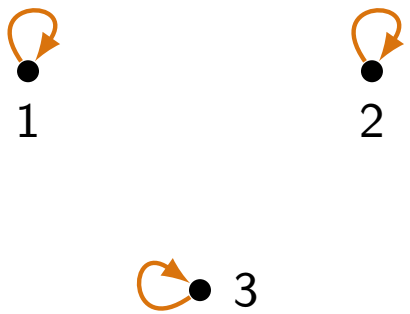
an element of X



$X = \{1, 2, 3\}$

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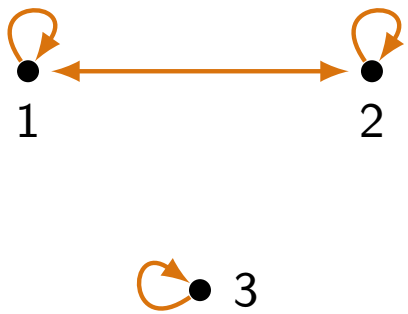


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dot	an element of X
loop at x	Rxx $\quad \langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle$

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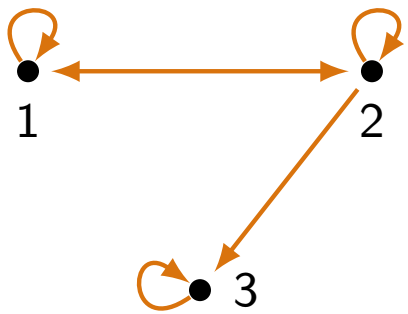


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two-way $x \leftrightarrow y$	Rxy *and* Ryx	$\langle 1,2\rangle, \langle 2,1\rangle$

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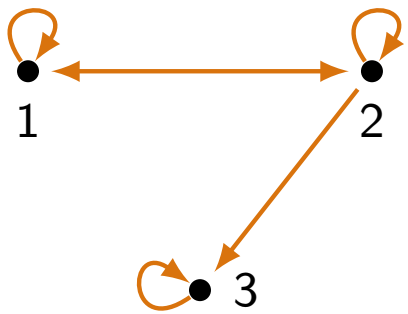


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one-way $x \rightarrow y$	Rxy only	$\langle 2,3\rangle$

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$X = \{1, 2, 3\}$

dot		an element of X
loop at x	Rxx	$\langle 1,1\rangle, \langle 2,2\rangle, \langle 3,3\rangle$
two-way $x \leftrightarrow y$	Rxy *and* Ryx	$\langle 1,2\rangle, \langle 2,1\rangle$
one-way $x \rightarrow y$	Rxy only	$\langle 2,3\rangle$

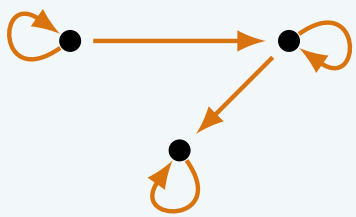
Next: properties a relation on X can have

Each property is a rule about the picture: *where are the loops? which arrows come back? which paths have shortcuts? Does it hold for all items?*

Reflexivity: where are the loops?

Reflexive: a loop on *every* element

Rxx for *all* $x \in X$

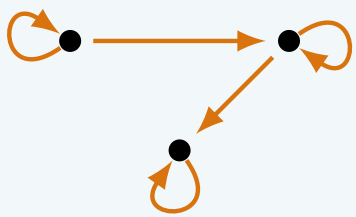


e.g., $=$, $\leq$, *is as old as*

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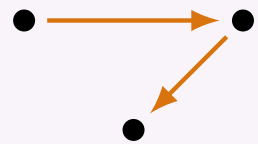
Rxx for *all* $x \in X$



e.g., $=$, $\leq$, *is as old as*

Irreflexive: *no* loops at all

Rxx for *no* $x \in X$

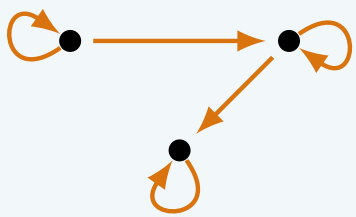


e.g., $<$, *is older than*, *is the mother of*

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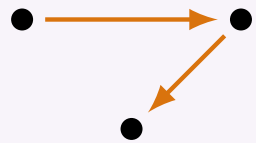
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Irreflexive: *no* loops at all

Rxx for *no* $x \in X$



e.g., $<$, *is older than*, *is the mother of*

Careful: these are *not* opposites

A relation can be *neither*: some elements have loops, others do not.

LOVES: j loves himself, m does not

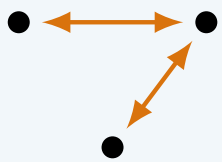


Symmetry: which arrows come back?

Symmetric

every arrow comes *back*

if Rxy then Ryx



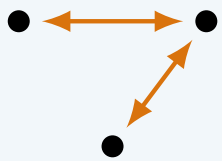
e.g., *is a sibling of*, *is married to*

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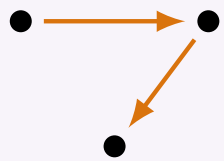


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Asymmetric

no arrow comes back

never both Rxy and Ryx



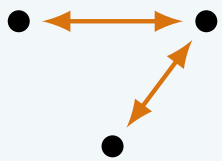
e.g., $<$, *is older than*
(so: no loops either!)

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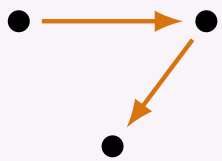


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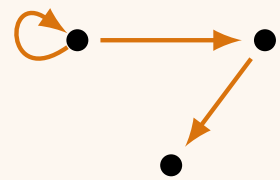


e.g., $<$, *is older than*
(so: no loops either!)

Anti-symmetric

only loops come back

if Rxy and Ryx then $x = y$



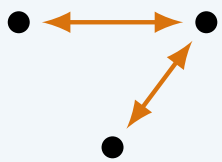
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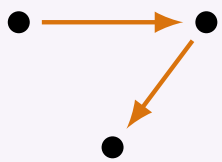


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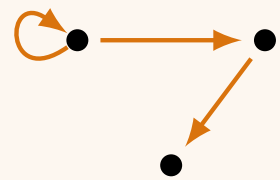


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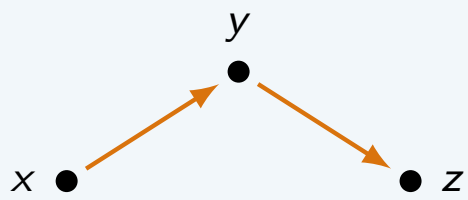
How they relate

asymmetric = anti-symmetric + irreflexive (a loop comes back to itself)

Transitivity: which paths have shortcuts?

Transitive: each path has a shortcut

if Rxy and Ryz, then also Rxz

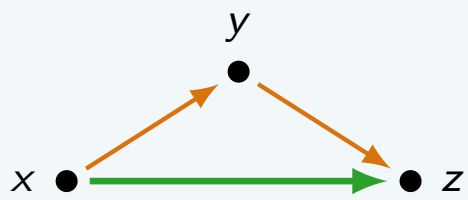


e.g., $<$, *is-older*, *is-ancestor*

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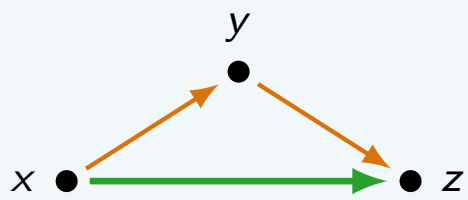


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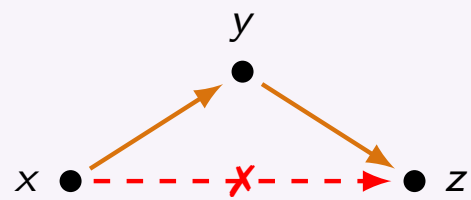
if *Rxy* and *Ryz*, then also *Rxz*



e.g., $<$, *is-older*, *is-ancestor*

Intransitive: *no* path has a shortcut

if *Rxy* and *Ryz*, then *not* *Rxz*

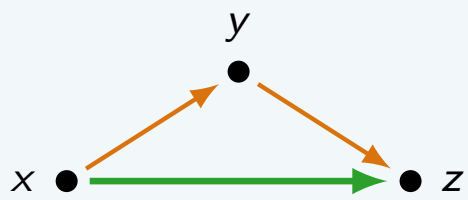


e.g., PRED, *is the mother of*

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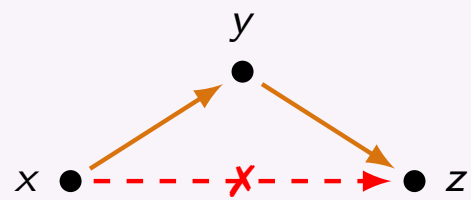
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e.g., PRED, *is the mother of*

Again: not opposites. Loves is *neither*

LOVES *js* and LOVES *sm*, but not LOVES *jm* (not transitive)
LOVES *jj* and LOVES *js*, and also LOVES *js* (not intransitive)

Connected: no two elements are strangers

Connected: any two *different* elements are linked

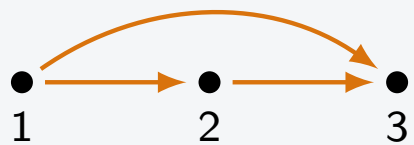
for all $x, y \in X$: Rxy or Ryx or $x = y$ in at least one direction

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Connected: any two *different* elements are linked

for all $x, y \in X$: Rxy or Ryx or $x = y$ in at least one direction

✓ **connected**



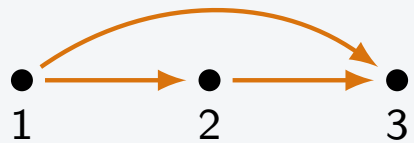
$<$ on $\{1, 2, 3\}$: every pair ordered

Connected: no two elements are strangers

Connected: any two *different* elements are linked

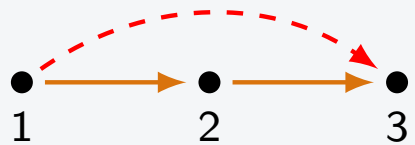
for all $x, y \in X$: Rxy or Ryx or $x = y$ in at least one direction

✓ **connected**



$<$ on $\{1, 2, 3\}$: every pair ordered

✗ **not connected**



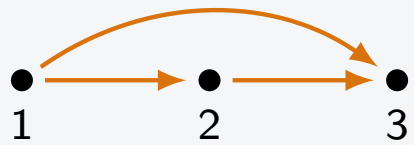
PRED: 0 and 2 are not linked

Connected: no two elements are strangers

Connected: any two *different* elements are linked

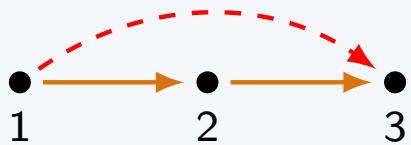
for all $x, y \in X$: Rxy or Ryx or $x = y$ in at least one direction

✓ **connected**



$<$ on $\{1, 2, 3\}$: every pair ordered

✗ **not connected**



PRED: 0 and 2 are not linked

Connected relations line everything up: think of a ranking, or the order of the numbers.

Practice: which properties does each relation have?

	refl.	irrefl.	sym.	asym.	anti.	trans.	intrans.	conn.
$=$ on $\mathbb{N}$	✓	✗	✓	✗	✓	✓	✗	✗
$<$ on $\mathbb{N}$	✗	✓	✗	✓	✓	✓	✗	✓
$\leq$ on $\mathbb{N}$	✓	✗	✗	✗	✓	✓	✗	✓
PRED on $\mathbb{N}$	✗	✓	✗	✓	✓	✗	✓	✗
is a sibling of	✗	✓	✓	✗	✗	✗*	✗	✗
LOVES	✗	✗	✗	✗	✗	✗	✗	✗

* Ann is a sibling of Bob and Bob of Ann, but Ann is not her own sibling.

Practice: which properties does each relation have?

Exercise (at home): validate the table

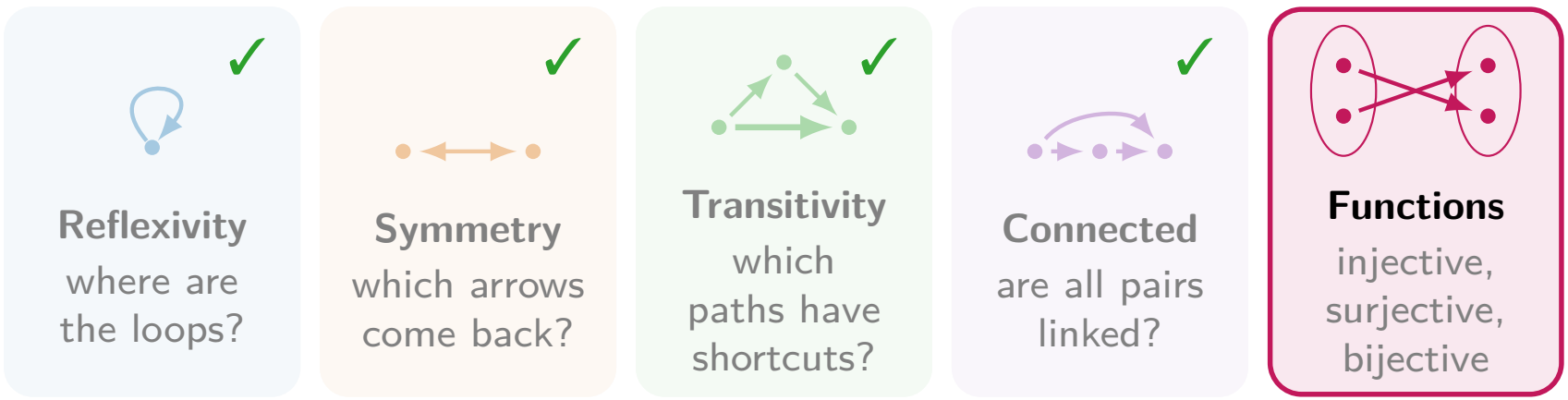
Check every ✓ and ✗ in the table yourself.

1. Draw a small diagram (a few elements are enough).
2. Check the picture: loops? arrows coming back? shortcuts?
3. For each ✗, find one counterexample.

Hint: LOVES is $\{\langle j, j\rangle, \langle j, s\rangle, \langle m, s\rangle, \langle s, m\rangle\}$.

Structural properties: where we are

Done: properties of relations. Next: properties of **functions**.



Properties of functions: arrows coming *into* Y

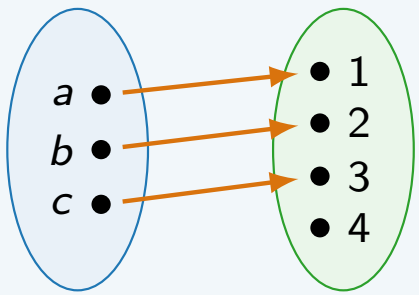
A function has one arrow *out of* each x. How many go *into* each y?

Properties of functions: arrows coming *into* Y

A function has one arrow *out of* each x. How many go *into* each y?

Injective

at most one arrow in $f(x_1) = f(x_2)$ implies $x_1 = x_2$



no shared outputs;
4 unused: fine

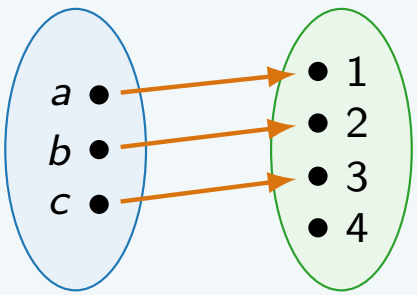
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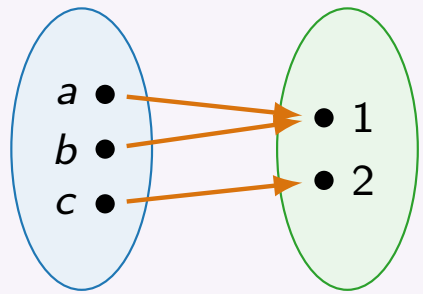


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Surjective

at least one arrow in

every y has an x with $f(x) = y$



range$(f) = Y$;
sharing: fine

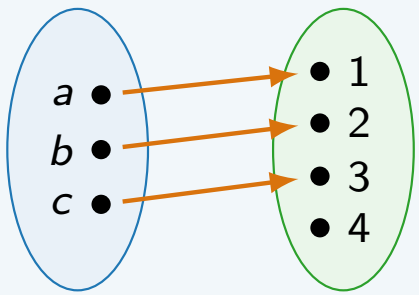
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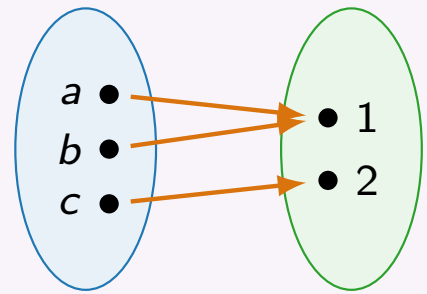


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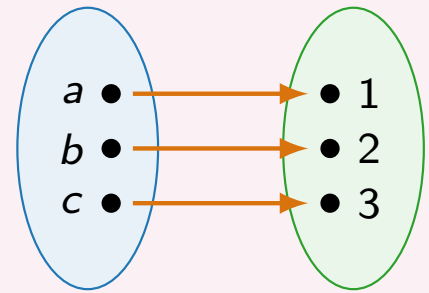


range$(f) = Y$;
sharing: fine

Bijective

exactly one arrow in

injective *and* surjective



a perfect pairing
of X and Y, both
injective/surjective.

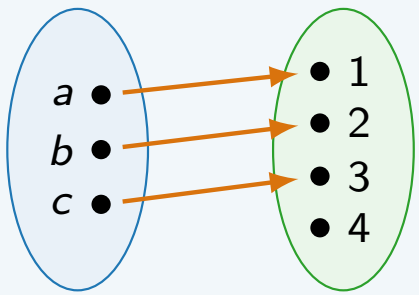
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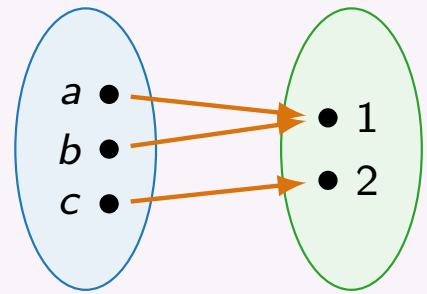


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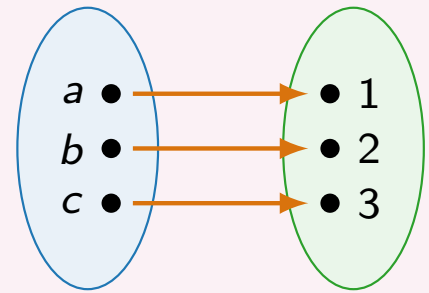


range$(f) = Y$;
sharing: fine

Bijective

exactly one arrow in

injective *and* surjective



a perfect pairing
of X and Y, both
injective/surjective.

Also called

an **injection** (one-to-one), a **surjection** (onto), a **bijection** (one-to-one correspondence)

Practice: injective, surjective, bijective?

	inj.	surj.	bij.	
$f(x) = x$ on $\mathbb{N}$	✓	✓	✓	the identity
$f(x) = x + 1$ on $\mathbb{N}$	✓	✗	✗	nothing maps to 0
$f(x) = x^2$ on $\mathbb{N}$	✓	✗	✗	nothing maps to 2
$f(x) = 0$ on $\mathbb{N}$	✗	✗	✗	$f(1) = f(2)$
CapitalOf countries $\to$ cities	✓	✗	✗	most cities aren't capitals

Practice: injective, surjective, bijective?

	inj.	surj.	bij.	
$f(x) = x$ on $\mathbb{N}$	✓	✓	✓	the identity
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$f(x) = 0$ on $\mathbb{N}$	✗	✗	✗	$f(1) = f(2)$
CapitalOf countries $\to$ cities	✓	✗	✗	most cities aren't capitals

Why care? Bijections compare sizes

$|X| = |Y|$ exactly when there is a bijection $f : X \to Y$ pair them up, nothing left over

Practice: injective, surjective, bijective?

	inj.	surj.	bij.	
$f(x) = x$ on $\mathbb{N}$	✓	✓	✓	the identity
$f(x) = x + 1$ on $\mathbb{N}$	✓	✗	✗	nothing maps to 0
$f(x) =$				
$f(x) =$				

Capi… …apitals

Why…

$|X| = |Y|$ exactly when there is a bijection $f : X \to Y$ — pair them up, nothing left over.

Exercise (at home): validate the table

For each function, draw a few arrows and check:

1. Do two inputs ever share an output? (injective?)
2. Is every output reached? (surjective?)
3. For each ✗, give one counterexample.

Conclusion: what you should know

Tuples and Cartesian products

- ✓ $\langle a, b \rangle \neq \langle b, a \rangle$: order matters
- ✓ $X \times Y$ and $|X \times Y| = |X| \cdot |Y|$

Conclusion: what you should know

Tuples and Cartesian products

- ✓ $\langle a, b\rangle \neq \langle b, a\rangle$: order matters
- ✓ $X \times Y$ and $|X \times Y| = |X| \cdot |Y|$

Relations

- ✓ $R \subseteq X \times Y$, written Rxy
- ✓ read and draw arrow diagrams
- ✓ relations on one set; n-ary
- ✓ $\mathrm{dom}(R)$ and $\mathrm{range}(R)$

Conclusion: what you should know

Tuples and Cartesian products

- ✓ $\langle a, b\rangle \neq \langle b, a\rangle$: order matters
- ✓ $X \times Y$ and $|X \times Y| = |X| \cdot |Y|$

Functions

- ✓ *exactly one* output per input
- ✓ codomain Y vs. range(f)

Relations

- ✓ $R \subseteq X \times Y$, written Rxy
- ✓ read and draw arrow diagrams
- ✓ relations on one set; n-ary
- ✓ dom(R) and range(R)

Conclusion: what you should know

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- ✓ reflexive, symmetric, transitive, . . .
- ✓ injective, surjective, bijective
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Next time: a bit about proofs, propositional logic.